

Quiz 12

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Given a matrix $A = \begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix}$. Find the eigenvalues λ 's and corresponding eigenvectors $\mathbf{x}$'s of A .

$$\begin{aligned}\det(A - \lambda I) &= \begin{vmatrix} 2-\lambda & 3 \\ 5 & 4-\lambda \end{vmatrix} \\ &= (2-\lambda)(4-\lambda) - 15 \\ &= 8 - 6\lambda + \lambda^2 - 15 \\ &= \lambda^2 - 6\lambda - 7 = (\lambda - 7)(\lambda + 1) = 0 \\ &\Rightarrow \lambda_1 = 7, \quad \lambda_2 = -1\end{aligned}$$

For $\lambda_1 = 7$, solve $(A - 7I)\mathbf{x} = 0$

$$\begin{aligned}\begin{bmatrix} -5 & 3 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= 0 \Rightarrow -5x_1 + 3x_2 = 0 \\ &\Rightarrow \mathbf{x} = c_1 \begin{bmatrix} 1 \\ \frac{5}{3} \end{bmatrix}\end{aligned}$$

For $\lambda_2 = -1$, solve $(A + I)\mathbf{x} = 0$

$$\begin{aligned}\begin{bmatrix} 3 & 3 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= 0 \Rightarrow 3x_1 + 3x_2 = 0 \\ &\Rightarrow \mathbf{x} = c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}\end{aligned}$$